ELEMENTAL BLEND SOLVER

Technical Reference & User Guide

Version 9.0.2 · Standalone HTML Application · No Install Required

73-Polymer Database · 20 Elements · 16 Integrated Analysis Tools
PCA Scatter · χ^2 Fit Test · Monte Carlo · Diagnostic Advisor · Solution Space Explorer

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Quick-Start Summary

The Elemental Blend Solver (EBS) is a single HTML file that runs in any modern browser with nothing to install. It finds the mass-mixing fractions of up to five source polymers whose combined elemental weight composition best matches a target molecule, using a constrained Non-Negative Least Squares (NNLS) solver.

System Requirements

Requirement	Details
Browser	Chrome ≥80, Edge ≥80, Firefox ≥78, Safari ≥14. Any Chromium-based browser (Brave, Opera, Vivaldi) also works.
Operating system	Windows, macOS, or Linux — runs identically on all three.
Internet connection	Not required. The application is entirely self-contained.
Mobile / tablet	Solver fully functional on iOS Safari ≥14. Note: Word and SVG export are unavailable on iOS (browser download restriction). Use desktop for exports.
File to download	molecular_solver_v9_0_2.html (one file, ~248 KB)

Three Steps to Your First Result

- Step 1 — Download: Go to https://github.com/pemason437-svg/Elemental-Blend-Solver and download molecular_solver_v9_0_2.html.
- Step 2 — Open: Double-click the file. It opens in your default browser. If nothing happens, right-click → Open with → [your browser].
- Step 3 — Read the result: The default problem (Choline Chloride target, five source polymers) is pre-loaded and solved. Fraction bars on the left show PVC 47.21%, POM 13.83%, Nylon 66 38.96%. The Diagnostic Advisor recommends next steps.

Status Bar at a Glance

Status	Meaning
✓ Exact solution	Target lies inside the convex hull. Blend is physically realisable. SSR < 1×10^{-8}.
≈ Noise-floor	RMSE < 0.5 pp and max gap < 1.0 pp. Residuals within instrument precision; treat as exact.
~ Acceptable fit	RMSE < 2.0 pp. Near-hull; small residuals. Review Blend Error Chart.
! Marginal fit	RMSE < 5.0 pp. Run Substitution Recommender or Exhaustive Search.
✗ Poor fit	RMSE ≥ 5.0 pp. Source set inadequate. Run Exhaustive Optimal Blend Search.

Where to Find Help

- New to the tool? Start with §4.6 (First Blend Walkthrough).
- Need the maths? See §2 (Mathematical Framework).
- Getting unexpected results? See §13 (Troubleshooting).
- Looking for a specific term? See §14 (Glossary).
- Bug reports and feature requests: https://github.com/pemason437-svg/Elemental-Blend-Solver/issues

1. Project Overview

EBS (molecular_solver_v9_0_2.html, ~248 KB) is a fully self-contained single-file HTML application that runs entirely in the browser — no installation, no server, no internet connection required. It computes the mass-mixing fractions of up to five source molecules whose combined elemental weight fractions best match a chosen target molecule across up to 20 tracked elements: C, H, O, N, Cl, S, P, Si, F, Br, I, Na, K, Mg, Ca, Fe, Al, B, Se, and Zn.

1.1 Key Features

- Live NNLS solver — results update instantly as any molecule changes
- Up to 5 source molecules; 8 target presets including Custom
- 73-polymer database covering 20 elements
- 16 integrated analysis tools: 6 quantitative chart views, 1 pedagogic Convergence Plot, and 9 panel diagnostics
- Diagnostic Advisor — integrates fit metrics into a single actionable recommendation
- 5-tier absolute RMSE classifier: Exact / Noise-floor / Acceptable / Marginal / Poor
- Foreign element detection — annotates elements absent from all sources
- 📄 Export Word Report — 8-section formal .docx with embedded SVG charts
- 🖼️ Export Charts (SVG) — all quantitative chart views as individual SVG files
- Light/Dark mode toggle (☀️/🌙) — preference saved across sessions

1.2 How to Run

- Download molecular_solver_v9_0_2.html to any folder
- Double-click to open in any modern browser (Chrome, Edge, Firefox, Safari)
- No npm, no Node.js, no VS Code, no dependencies required

2. Mathematical Framework

2.1 Weight Fraction Definition

For a molecule with atom counts $\{n_i\}$ over $p \leq 20$ active elements, the molecular weight and elemental weight fraction of element X are:

$$M = \Sigma\, AW(e_i) \cdot n_i \quad (g/mol) \qquad [Eq.\ 1]$$

$$w_X = AW(X) \times count(X) / M \qquad [Eq.\ 2]$$

All elemental weight fractions for a molecule sum to 1.0. EBS uses IUPAC 2021 standard atomic weights (Prohaska et al. 2022). Selected values: C=12.011, H=1.008, O=15.999, N=14.007, Cl=35.45, S=32.06, P=30.974, Si=28.085, F=18.998, Br=79.904, I=126.904, Na=22.990, K=39.098, Mg=24.305, Ca=40.078, Fe=55.845, Al=26.982, B=10.811, Se=78.971, Zn=65.38 g/mol.

2.2 The Linear System

Let f_i be the mass fraction of source polymer i in the blend. The achieved elemental composition is:

$$A \cdot f = b \qquad [Eq.\ 3]$$

where $A \in \mathbb{R}^{m \times n}$ is the (elements × sources) weight-fraction matrix, $b \in \mathbb{R}^m$ is the target elemental weight-fraction vector, and $f \in \mathbb{R}^n$ is the mixing-fraction vector subject to $f \geq 0$ and $\Sigma f_i = 1$. The matrix A has up to 20 rows (one per tracked element) and up to 5 columns (one per source).

2.3 Solution Strategy — Unified NNLS

EBS uses a single unified Non-Negative Least Squares (NNLS) solver via the Lawson-Hanson active-set algorithm for all system sizes. The sum constraint ($\Sigma f_i = 1$) is incorporated as a heavily weighted extra row appended to A and b (penalty weight $\lambda = 10$):

$$\bar{A} = [A\ ;\ \lambda \cdot 1^T], \quad \bar{b} = [b\ ;\ \lambda] \qquad [Eq.\ 4]$$

This converts the equality-constrained problem to standard NNLS form. Post-solve, raw weights are renormalised so $\Sigma f_i = 1$ exactly ($|\Sigma f_i - 1| < 10^{-8}$ after renormalisation). Sensitivity analysis confirms SSR and mixing fractions are stable across $\lambda \in [5, 100]$; fractions vary by less than 0.1 pp across this range.

System Size	System Type	Method
n < 5	Underdetermined	NNLS with weighted sum constraint row
n = 5	Square	NNLS (exact solution if target in hull)
n > 5	Overdetermined	NNLS — constrained least-squares

2.4 Goodness-of-Fit Assessment

Solution quality is reported via four complementary metrics:

- SSR = $||A\hat{f} - b||^2$: Primary goodness-of-fit metric. SSR < 1×10^{-8} indicates an exact solution.
- RMSE = $\sqrt{(SSR/m)} \times 100$ pp: Root mean squared error across active elements, in percentage points.
- $\chi^2 = \Sigma(\text{achieved}_e - \text{target}_e)^2/\sigma_e^2$: Chi-squared statistic. σ(C,H,N)=0.003, σ(O,S,P,Si,Na,K,Mg,Ca,Fe,Al,Zn,B)=0.004, σ(Cl,F,Br,I,Se)=0.005 (weight fractions). Degrees of freedom df = m − nactive (clamped to ≥1). P-values via Wilson-Hilferty normal approximation.
- GDS (Global Decomposability Score) = $\max(0, 1 - RMSE/\|b\|_2)$: Dimensionless [0,1] scalar; higher is better.

The 5-tier absolute RMSE classifier provides the primary pass/fail decision, grounded in standard CHN/XRF instrument noise (σ = 0.3–0.5 pp):

Tier	Condition	Action
✓ Exact	SSR < 1×10^{-8}	Genuine hull-interior reconstruction. Accept fractions.
≈ Noise-floor	RMSE < 0.5 pp AND $\max\|r_i\|$ < 1.0 pp	Within single-σ instrument noise. Treat as exact.
~ Acceptable	RMSE < 2.0 pp	Near-hull; small residuals. Review Blend Error Chart.
! Marginal	RMSE < 5.0 pp	Run Substitution Recommender.
✗ Poor	RMSE ≥ 5.0 pp	Run Exhaustive Optimal Blend Search.

Note on χ^2 conflicts: When the tier is Acceptable but χ^2 is Poor, this typically reflects a structurally false null hypothesis (the target lies outside the source hull). The Diagnostic Advisor (§7) resolves this conflict automatically.

2.5 Infeasibility Detection

When only one source supplies a given element, its fraction is fully determined ("pinned") by that element's balance:

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f_source = w_element,target / w_element,source    [Eq. 5]
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If the sum of all pinned fractions exceeds 1.0, no physical blend exists — this is a mathematical proof of infeasibility. The solver still returns the best-fit approximation. Example (Choline Chloride,

default blend): PVC is the only Cl source → f_PVC = 44.76%. Nylon 66 is the only N source → f_Nylon = 81.05%. Sum = 125.8% > 100% → proven infeasible.

3. Reference Problem: Choline Chloride Target

3.1 Target Molecule

Molecule	Formula	MW (g/mol)	Elemental Weight %
Choline Chloride	$C_5H_{14}NOCl$	139.63	C 43.01% H 10.11% O 11.46% N 10.03% Cl 25.39%

3.2 Source Molecules and Weight Fractions

Polymer	Formula	MW	%C	%H	%O	%N	%Cl
PVC	C_2H_3Cl	62.50	38.44	4.84	—	—	56.73
PET	$C_{10}H_8O_4$	192.17	62.50	4.20	33.30	—	—
Polypropylene	C_3H_6	42.08	85.63	14.37	—	—	—
POM (Acetal)	CH_2O	30.03	40.00	6.71	53.29	—	—
Nylon 66	$C_{12}H_{22}N_2O_2$	226.32	63.69	9.80	14.14	12.38	—

3.3 Default Solve Result

EBS returns: PVC 47.21% / POM 13.83% / Nylon 66 38.96% / PET 0.00% / Polypropylene 0.00%. SSR = 7.056×10^{-3}; RMSE = 3.76 pp (Marginal tier); GDS = 0.929. No exact physical blend of these five polymers can replicate Choline Chloride's elemental profile — the pinned fractions sum to 125.8% > 100%. The Exhaustive Optimal Blend Search recovers GDS ≈ 0.975 with a different five-polymer combination (HDPE/PVDC/Urea/Glycine, SSR ≈ 9.166×10^{-4}).

4. Running the Application

4.1 Opening the App

- Download molecular_solver_v9_0_2.html from the project repository: https://github.com/pemason437-svg/Elemental-Blend-Solver
- Double-click the file — it opens in your default browser
- The solver is live immediately; the default problem (Choline Chloride target, 5 source polymers) is pre-loaded

4.2 Changing the Target Molecule

Click the blue Target Molecule card at the top left. A modal will open where you can:

- Pick a preset: Choline Chloride, Glucose, Urea, Glycine, Alanine, Adipic Acid, Caprolactam, or Custom
- Type any name and molecular formula using any of the 20 supported elements (e.g. C6H12O6 for Glucose)
- Preview the molecular weight and weight fractions before saving

4.3 Editing Source Molecules

- Click "edit" on any source card to rename it or change its formula
- Click × to remove a source
- "+ Add molecule" to enter a custom source (up to 5 total)
- "From library" to add from the 15 preset common plastics

4.4 Reading the Results

Panel	Contents
Fraction bars	Each source card shows a colour bar with its mass mixing fraction percentage, updating live.
Elemental comparison	Table comparing target vs achieved weight % for each element, with Δ (pp) and colour-coded status (green <0.5 pp, amber <2 pp, red >2 pp).
Status bar	Shows fit tier (✓ Exact / ≈ Noise-floor / ~ Acceptable / ! Marginal / ✗ Poor) with RMSE and max error.
Diagnostic Advisor	Integrates GDS tier and χ^2 into a single actionable recommendation (§7). Always review this first.
χ^2/GDS chip	Reports χ^2, df, p-value, and 5-tier RMSE classification. GDS scalar shown for trend monitoring.
Graphical analysis	Six quantitative SVG chart views (§6) plus pedagogic Convergence Plot.

Panel	Contents
Advanced tools	Nine quantitative panel diagnostics (§8) including Solution Space Explorer.

4.5 Export Buttons

Button	Action
📄 Export Word Report	Generates a complete formal .docx report: cover page, 8 sections, all charts embedded as SVG. Zero external dependencies — fully offline.
🖼️ Export Charts (SVG)	Downloads all quantitative chart views as individual .svg files. Opens in any browser, Inkscape, Illustrator, or inserts into Word/PowerPoint.

Note: On iOS Safari, the solver and all analysis tools function fully. Word and SVG export are unavailable due to iOS browser download restrictions. Use a desktop browser for exports.

4.6 First Blend Walkthrough — Glucose from Simple Sources

This walkthrough demonstrates an exact solution. Glucose ($C_6H_{12}O_6$) can be matched exactly by POM Acetal (CH_2O) alone, since both share the CH_2O empirical ratio.

- Step 1 — Set target: Click the Target Molecule card → select preset "Glucose" → Save. Card shows MW = 180.156 g/mol, C 40.00%, H 6.71%, O 53.29%.
- Step 2 — Clear sources: Click × on each of the five default sources to remove them all.
- Step 3 — Add Polypropylene: Click + New molecule → select preset "Polypropylene" → Save.
- Step 4 — Add POM (Acetal): Click + New molecule → select preset "POM (Acetal)" → Save. Notice POM's elemental profile is identical to Glucose.
- Step 5 — Read result: Status bar shows "✓ Exact solution". Fraction bars: POM 100.000%, Polypropylene 0.000%. SSR < 1×10^{-8}; RMSE ≈ 0.0000%.
- Step 6 — Explore: Try the Substitution Recommender (§8) and Exhaustive Search (§8) to find alternative source combinations for other targets.

5. Presets Reference

5.1 Target Molecule Presets

Target	Formula	Notes
Choline Chloride	$C_5H_{14}NOCl$	Default target. Requires Cl and N sources. Provably infeasible with default 5-polymer blend.
Glucose	$C_6H_{12}O_6$	Simple carbohydrate. C, H, O only — matched exactly by POM alone.
Urea	CH_4N_2O	High N content. Good NNLS test case.
Glycine	$C_2H_5NO_2$	Simplest amino acid. Low MW, moderate N and O.
Alanine	$C_3H_7NO_2$	Aliphatic amino acid. Similar to Glycine.
Adipic Acid	$C_6H_{10}O_4$	Nylon 66 monomer. C, H, O only.
Caprolactam	$C_6H_{11}NO$	Nylon 6 monomer. Contains N.
Custom	—	Enter any formula using any of the 20 supported elements.

5.2 Source Polymer Presets (15 built-in)

Name	Formula	MW	Key elements
PVC	C_2H_3Cl	62.50	Only Cl source in default blend
PET	$C_{10}H_8O_4$	192.17	High O, moderate C/H
Polypropylene	C_3H_6	42.08	Pure hydrocarbon — C and H only
POM (Acetal)	CH_2O	30.03	Very high O fraction (~53%)
Nylon 66	$C_{12}H_{22}N_2O_2$	226.32	Only N source in default blend
Polycarbonate	$C_{16}H_{14}O_3$	254.28	Aromatic, moderate O
HDPE	C_2H_4	28.05	Pure hydrocarbon — lowest MW source
Polystyrene	C_8H_8	104.15	Aromatic, C/H only
PMMA (Acrylic)	$C_5H_8O_2$	100.12	Moderate O (~32%)
Nylon 6	$C_6H_{11}NO$	113.16	N-containing, good H/N source
PLA	$C_3H_4O_2$	72.06	High O, biodegradable polyester
PVDC	$C_2H_2Cl_2$	96.94	High Cl (~73%) — strongest Cl source

Name	Formula	MW	Key elements
PAN	C_3H_3N	53.06	High N (~26%) — strong N source
PCL	$C_6H_{10}O_2$	114.14	Moderate O, no N/Cl
PVAc	$C_4H_6O_2$	86.09	Moderate O (~37%)

5.3 Extended Polymer Database (73 entries, Advanced Tools only)

The advanced analytical tools search across a 73-polymer database covering all 20 tracked elements. The database spans 8 chemical families: chlorinated polymers, nitrogen polymers, oxygenated polymers, hydrocarbon polymers, biopolymers, specialty polymers (Si, P, F, S), metal-containing entries (Na, B, Br, I, K, Mg, Ca, Fe, Al, Se, Zn), and monomers/small molecules (Urea, Glycine, Alanine, Caprolactam, DOP/DINP plasticizers). All 73 formulas are sourced to IUPAC atomic weights and verified against the Polymer Handbook and CRC Handbook.

For a given target, database entries carrying elements entirely absent from the target ("foreign elements") are excluded before Exhaustive Search runs. For the Choline Chloride target, 18 entries are excluded, leaving a pool of 55 eligible entries — C(55,5) = 3,478,761 combinations.

6. Graphical Analysis — Seven Views

EBS provides six quantitative interactive chart views plus one pedagogic visualization. All charts update live whenever any molecule is changed and reflect the exact current solver state. Click tab buttons to switch views. The SVG background is forced to white on export for clean printing.

6.1 View Descriptions

View	Chart Type	What it shows
1 — Radar	Spider / polygon	All tracked elements as radial axes. White polygon = target; dashed coloured polygons = each source at its computed fraction; purple = blend. Overlap confirms close elemental matching.
2 — PCA Scatter	2D projection + convex hull	Target and sources projected onto PC1–PC2 of the active-source covariance matrix. Dashed hull = active sources; grey dots = all 73 database entries. Gap between target star and hull edge shows Distance-to-Hull visually. Grey dots between star and hull edge are productive substitution candidates.
3 — Build-Up	Stacked bar	Each bar shows one source's elemental contribution at its computed fraction. "Blend" column sums all contributions. "Target" column is the reference. Gaps are residual errors.
4 — Blend Error	Diverging bar	Signed error per element (achieved − target) in pp. Right = over-supply; left = under-supply. Colour: green <0.5 pp, amber <2 pp, red >2 pp.
5 — Feasibility	Animated flow diagram	Step-by-step proof of feasibility or infeasibility. Each step reveals one constraint: single-source elements pin fractions; their sum is checked against 100%; optimiser statement shown. Auto-animates or step manually.
6 — Composition	Grouped bar	Side-by-side target (blue) vs achieved (orange) elemental weight fractions. Foreign elements absent from all sources appear as unmatched blue bars, each annotated with residual Δ in pp and the label "Not in sources".
Convergence Plot (pedagogic)	Log-scale line	Illustrates projected-gradient SSR descent for teaching purposes only. This is NOT a trace of the actual Lawson-Hanson solver. Do not cite as solver performance data. For rigorous convergence diagnostics use Monte Carlo and Leave-One-Out.

6.2 Chart Export

- Individually via "🖼️ Export Charts (SVG)": downloads each quantitative chart as a standalone SVG file.

- All embedded in the Word report via "📄 Export Word Report": SVG files written directly into docx word/media/ folder, requiring no rasterisation.

Charts are serialised directly as SVG XML strings using XMLSerializer(), with an explicit white background rect injected. This approach avoids the canvas/PNG pathway which modern browsers block for SVG blob URLs (cross-origin security restriction).

7. Diagnostic Advisor

The Diagnostic Advisor is a result-panel element displayed immediately after every solve. It integrates the GDS tier and χ^2 outcome into a single actionable recommendation, eliminating ambiguity when the two fit metrics disagree. The Advisor is hidden when no source materials are loaded.

The Advisor evaluates seven mutually exclusive conditions in order of precedence:

Condition	Panel colour	Recommendation
Exact (SSR < 10^{-8})	Green	Run Monte Carlo Sensitivity Analysis to confirm the exact solution is well-constrained.
Wrong sources — exact match exists elsewhere (amber global status)	Amber	Run Substitution Recommender. A single source swap will in most cases close the gap.
Noise-floor (RMSE < 0.5 pp)	Green	Analytically equivalent to exact. Run Monte Carlo to verify CI stability.
Acceptable — no χ^2 conflict	Amber	Run Monte Carlo then inspect Blend Error Chart.
Acceptable — χ^2 conflict (Acceptable tier but χ^2 is Poor)	Amber	Advisor runs a lightweight best-swap scan. If best swap reduces SSR >10%, recommends Substitution Recommender with estimated SSR reduction. Otherwise recommends Exhaustive Search. An inline note explains the structural cause of the conflict.
Marginal (RMSE < 5.0 pp)	Orange	Run Substitution Recommender. Guidance on element with largest residual shown.
Poor (RMSE ≥ 5.0 pp)	Red	Run Exhaustive Optimal Blend Search. Worst-gap element flagged when a foreign element is detected.

Validation: all seven Advisor branches were tested across 90 cases (30 molecules × 3 source sets). The 5-tier RMSE classifier achieves 0% false-positive rate and 0% false-negative rate across the full validation suite.

8. Advanced Diagnostic Tools

Nine quantitative panel diagnostics run automatically after each solve, together with the six quantitative chart views and the Convergence Plot, giving 16 integrated analysis tools in total. Each addresses a distinct decision question that cannot be answered by SSR or mass-fraction output alone.

8.1 Solution Space Explorer (new in v9)

Characterises the algebraic solution space of exact or near-exact blends. The dimensionality dim = n − rank(A) determines uniqueness:

- dim = 0: Unique solution. A source-exclusion panel identifies database polymers that can substitute for each active source at SSR ≈ 0.
- dim > 0: Valid blends form a polytope. The tool reports per-source fraction ranges across its vertices — essential for practitioners who must choose among equivalent blends on the basis of cost or availability.

This tool is complementary to Monte Carlo (noise-driven uncertainty) and Leave-One-Out (single-source redundancy). It answers: are there other blends that are exactly equivalent to the NNLS solution?

8.2 Sensitivity Analysis — Monte Carlo Perturbation

Propagates source-composition uncertainty to mixing-fraction confidence intervals via 2,000 NNLS replicates with multiplicative Gaussian noise (σ = 2% of each elemental fraction). The 2% level models repeat-unit formula approximation error relative to commercial-grade polymers. For moisture-sensitive polyesters, consider σ = 5–10%.

Statistic	Symbol	Interpretation
Mean	μ	Expected value of fraction or SSR across all perturbations
Standard deviation	σ	Spread of the distribution; larger σ = more sensitive to input uncertainty
5th–95th percentile	P5–P95	90% confidence interval
CI width (P95−P5)	ΔCI	Key stability metric: <4% = stable, 4–8% = moderate, >8% = sensitive

Output: CI strip chart (wide band = 90% CI, narrow = IQR, tick = median, white marker = base solution). A CI half-width exceeding the fraction itself indicates a poorly constrained source that should be reported as a range rather than a point estimate. For the Choline Chloride default: PVC and Nylon 66 have moderate CIs (pinned by element constraints); PET and POM have wide CIs.

8.3 Distance-to-Hull Metric

Computes $d = \sqrt{SSR} = \min f \|Af - b\|_2$, the L_2 distance from the target to the nearest achievable blend on the source convex hull. Three action tiers:

- $d \approx 0$: Exact — accept fractions directly.
- $0 < d \leq 0.01$: Within the instrument noise floor (~0.005–0.010 for a five-element analysis) — accept with Monte Carlo confirmation.
- $d > 0.01$: Actionable infeasibility — run Substitution Recommender or Exhaustive Search.

Also computes per-element signed gap (target − nearest), binding constraint certificate (e.g. "Target needs 8.5% more H than the hull can provide"), and supplement candidates scored by cosine similarity between elemental profiles and the gap vector.

8.4 Polymer Substitution Recommender

Tests all eligible database polymers as single-slot replacements across all 5 source slots and returns a ranked table by SSR. For the Choline Chloride case (55 eligible entries, 5 slots), this replaces up to 275 manual re-runs. A top substitution reducing SSR by >50% warrants adoption; if improvement is <10%, run Exhaustive Search.

Strategy: (1) slot-wise replacement — each database polymer tried in each slot, solver re-run; (2) deficit-guided scoring — O(n) operation scoring polymers by dot product of their elemental profile with the positive components of the current deficit vector. Both strategies are combined for the final ranked output.

8.5 Leave-One-Out Source Importance Analysis

Re-solves with each source removed in turn; measures $\Delta SSR^i = SSR(\text{without source } i) - SSR(\text{baseline})$. Three calibrated thresholds based on the 20-molecule validation suite:

- Indispensable: $\Delta SSR/SSR_full > 10^3$ — sole elemental carrier, irreplaceable.
- Informative: $\Delta SSR/SSR_full$ in $(1.0, 10^3)$ — meaningful contribution but not sole carrier.
- Near-redundant: $\Delta SSR/SSR_full < 1.0$ — removing increases SSR by less than 100%; candidate for substitution.

When a source is near-redundant because another source has an identical elemental profile, the output is annotated "compositional duplicate of [name]". For Choline Chloride default: PVC and Nylon 66 are Indispensable; POM is Informative; PET and Polypropylene are Near-redundant (0% fraction).

8.6 Global Decomposability Score (GDS)

$GDS = \max(0, 1 - RMSE/\|b\|_2)$, where $RMSE = \sqrt{(SSR/m)}$. Answers the meta-question: is this target decomposable from any combination in the full 73-polymer database? Solves NNLS with all 73 database polymers simultaneously. Output: global SSR; binding element; top contributors in the globally optimal blend; verdict (green = inside global hull; amber = outside).

For Choline Chloride: global SSR ≈ 1.8×10^{-3} (near-decomposable). Top contributors: PVC (~40%), Urea (~22%), Nylon 66 (~21%), POM (~12%), Chitosan (~5%).

8.7 PCA Database Coverage Map

Projects all 73 database entries, nominated sources, and the target onto 2D PCA score space. The shaded hull encloses all five nominated sources (including zero-fraction ones), representing the maximum accessible blend space. Target marker carries a three-state colour code from a global 73-material NNLS scan: green (exact match possible), amber (exact match exists in database but not with current sources), red (no database combination can match — truly infeasible). Use this map to identify database polymers near the target that are not in the current selection.

8.8 Ternary SSR Contour Diagram

Available when exactly 3 sources are loaded. The full 3-source simplex (triangle) is sampled at 55×55 resolution; SSR is evaluated for each grid point and displayed as a green→red colour contour. A tight low-SSR region indicates a well-defined, robust optimum; a broad flat region indicates that the data do not distinguish between many blend recipes. The NNLS-optimal point is marked with a white circle. Colour legend shows 2×, 5×, and 10× minimum SSR reference lines.

8.9 Exhaustive Optimal Blend Search

Guarantees the globally optimal r-polymer blend from the eligible database by evaluating every C(pool, r) combination. An inline Web Worker is spun up (no server required). A live progress bar shows combinations processed and current best SSR. On completion, the top-10 globally optimal blends are displayed ranked by SSR.

n	C(55,n)	NNLS calls	Expected time	Notes
2	1,485	1,485	< 0.1 s	Instant; good first pass
3	26,235	26,235	< 1 s	Effective for most analyses
4	341,055	341,055	~7 min	Reserve for unresolved targets
5	3,478,761	3,478,761	~58 min	Global guarantee; run overnight

Combination counts assume 18 foreign-element exclusions for the Choline Chloride target (pool = 55). Pool size varies by target; C(73,5) = 15,020,334 when no entries are excluded.

9. Word Report Export

Click "📄 Export Word Report" to generate a complete formal .docx report. The entire report is built client-side using raw OOXML and a minimal ZIP builder in pure JavaScript — no CDN, no library dependencies, fully offline.

Section	Title	Contents
Cover	Title Page	Target molecule name and formula, solution quality badge, RMSE/SSR/GDS, generation date
1	Executive Summary	Solution overview; convex combination explanation; conservation of mass
2	Problem Setup	Target elemental fractions table; all source molecules with weight-fraction profiles and MW
3	Optimal Mixing Fractions	Mass fraction table with per-100g column and plain-English blend instruction
4	Elemental Comparison	Target vs achieved per element, Δ (pp), relative error, colour-coded status
5	Feasibility Analysis	Pinned fraction derivations; infeasibility sum check; hull distance metric
6	Graphical Analysis	All quantitative chart views embedded as SVG images with descriptive captions
7	Sensitivity & Advanced	Monte Carlo statistics table; supplement candidates; top-8 substitution recommendations
8	Methodology	Full NNLS mathematical formulation; Lawson-Hanson algorithm description; atomic weights reference table

10. Technical Methodology

10.1 Full Mathematical Formulation

Let n be the number of source polymers and $m \leq 20$ be the number of active elements. Define $A \in \mathbb{R}^{m \times n}$ as the weight-fraction matrix, $b \in \mathbb{R}^m$ as the target elemental weight-fraction vector, and $f \in \mathbb{R}^n$ as the mixing-fraction vector. The constrained optimisation problem is:

$$\text{minimise } ||Af - b||^2 \quad \text{subject to: } f \geq 0, \quad \Sigma f_i = 1 \qquad \text{[Eq. 6]}$$

The sum constraint is converted to an unconstrained NNLS problem by appending a penalty row ($\lambda = 10$) to A and b, giving the augmented system $\bar{A}\bar{f} = \bar{b}$. The Lawson-Hanson algorithm solves the augmented NNLS problem to a global minimum. Post-solve renormalisation corrects fractions to sum to exactly 1.000 ($|\Sigma f_i^* - 1| < 10^{-8}$).

10.2 Lawson-Hanson Active-Set Algorithm

The NNLS algorithm maintains a passive set P of "active" (non-zero) variables. At each iteration: compute the gradient; if all inactive variables have non-positive gradient components, the KKT conditions are satisfied — stop; otherwise, move the most violating variable into P; solve the reduced unconstrained LS system for passive variables; if any variable goes non-positive, perform a ratio test and remove it, then repeat. Convergence is guaranteed in a finite number of steps (bounded by 2^n in the worst case; typically $O(n^2)$ for polymer systems). The inner-loop interpolation safeguard (alpha-step line search) prevents cycling.

10.3 Elemental Properties

Element	Symbol	AW (g/mol)	Role in typical polymers
Carbon	C	12.011	Backbone of all organic polymers
Hydrogen	H	1.008	Saturating bond valences; varies widely with unsaturation
Oxygen	O	15.999	Esters, ethers, carbonyls; absent in pure hydrocarbons
Nitrogen	N	14.007	Amides (nylons), nitriles (PAN), ureas; relatively rare
Chlorine	Cl	35.45	PVC, PVDC, Epoxy; high MW contribution from single atom
Sulfur	S	32.06	Vulcanised rubbers, polysulfone
Phosphorus	P	30.974	Flame retardants, phosphate esters
Silicon	Si	28.085	Silicones (PDMS), glass-filled composites
Fluorine	F	18.998	PTFE, PVDF, fluoropolymers; σ=0.005 in χ^2 test
Bromine	Br	79.904	Brominated flame retardants

Element	Symbol	AW (g/mol)	Role in typical polymers
Iodine	I	126.904	Poly(4-iodostyrene), iodinated contrast agents
Sodium	Na	22.990	Na-CMC, sodium acrylate ionomers
Potassium	K	39.098	Potassium polyacrylate (superabsorbent)
Magnesium	Mg	24.305	Magnesium stearate (lubricant/stabiliser)
Calcium	Ca	40.078	Calcium stearate (heat stabiliser in PVC)
Iron	Fe	55.845	Iron oxide pigments, iron stearate lubricants
Aluminium	Al	26.982	ATH flame retardant (AlH_3O_3), aluminium stearate
Boron	B	10.811	Hexagonal boron nitride (h-BN) ceramic filler
Selenium	Se	78.971	Selenomethionine, selenium-containing biochemicals; σ=0.005
Zinc	Zn	65.38	Zinc stearate (lubricant/heat stabiliser)

11. Limitations

EBS addresses elemental composition matching only. Four categories of physical, chemical, mathematical, and analytical limitations apply. High-priority items for the next development cycle are listed below.

11.1 Physical Limitations

- Bulk and melt density are not modelled. PVC (~1.38 g/cm³) and Nylon 66 (~1.14 g/cm³) differ substantially in density from small-molecule targets such as Choline Chloride (~1.07 g/cm³). Elemental matching is a necessary but insufficient condition for practical equivalence. Future version: density field per source, blend density prediction.
- Polymer miscibility and phase separation are not addressed. Most polymer pairs are thermodynamically immiscible. A blend may be elementally optimal while phase-separating in practice. Future version: binary compatibility table, Flory-Huggins χ parameter estimates.
- Thermal processing windows are not considered. POM depolymerises at ~200°C; Nylon 66 requires 260–280°C — a fundamental incompatibility. Future version: Tg, Tm, Td fields; feasible processing window analysis.
- Molecular weight distribution and chain architecture are not represented. Melt viscosity scales as Mw^3.4; two PET batches with different Mw can have viscosities differing by an order of magnitude. Future version: MW, PDI, and viscosity estimate fields.

11.2 Chemical and Analytical Limitations

- Inorganic fillers, plasticisers, and additives are not represented. Flexible PVC containing 30% DOP plasticiser has an effective carbon content of ~45%, not 38.4% from the pure repeat unit. Future version: compound source compositions as mixtures of formulas.
- Target measurement uncertainty is not propagated. When the target is derived from elemental analysis rather than a known formula, CHN analysers carry ±0.3 pp accuracy per element; current sensitivity analysis only accounts for source uncertainty. Future version: per-element target uncertainty field, joint Monte Carlo.
- The solver assumes exact stoichiometry of repeat units. Copolymer compositions vary between batches. Future version: composition variability flag and range per database entry.
- Structural isomers are indistinguishable. Polymers sharing the same empirical repeat-unit formula cannot be separated by elemental data alone. Confirm blend identity by FTIR or NMR.

11.3 Mathematical Limitations

- The solution landscape is not fully explored. NNLS returns one point in the feasibility polytope; for underdetermined systems there may be a continuum of equally valid solutions. The Solution Space Explorer (§8.1) reports per-source fraction ranges for exact solutions. Future version: full polytope vertex enumeration for approximate solutions.
- No multi-objective optimisation. The solver optimises elemental SSR only; cost, density, or Tg trade-offs are not modelled. Future version: weighted multi-objective with Pareto front.

- The Convergence Plot depicts projected gradient, not Lawson-Hanson. The chart is illustrative only. Do not cite as solver performance data.
- Chi-squared degrees of freedom formula. df = m − nactive is exact for interior-active solutions; for boundary solutions it is a conservative lower bound that may slightly inflate χ^2. When df ≤ 2, treat p-values as approximate indicators only.

12. Software Verification and Validation

EBS v9.0.2 was validated across three independent activities: (1) a 20-molecule numerical test suite against a Python/scipy reference implementation; (2) an expanded 90-case computational study (30 molecules × 3 source sets) for GDS classifier validation; and (3) a computational self-consistency check against published CHNS measurements from the BOTTLE Plastic Substrates Database.

12.1 20-Molecule Numerical Suite

Two tiers: Tier 1 — 12 synthetic targets constructed by randomly sampling blend fractions over the five default sources (exact recovery guaranteed by construction; these verify numerical precision only). Tier 2 — 8 real-molecule targets providing independent validation across four categories: hull-interior (expected SSR=0), near-boundary, outside-hull (best-approximation), and foreign-element.

Selected results:

Target	Formula	Hull Status	SSR	GDS Tier	Dominant Sources
Choline Chloride	$C_5H_{14}NOCl$	Outside	7.056×10^{-3}	Marginal	PVC 47%, Nylon 66 39%
Glucose	$C_6H_{12}O_6$	Interior	$\leq10^{-10}$	Exact	POM 100%
Adipic Acid	$C_6H_{10}O_4$	Interior	$\leq10^{-10}$	Exact	PET 22%, PP 10%, POM 68%
Caprolactam	$C_6H_{11}NO$	Interior	$\leq10^{-10}$	Exact	Nylon 66 100%
Urea	CH_4N_2O	Outside	2.663×10^{-1}	Poor	POM 48%, Nylon 66 42%

12.2 90-Case GDS Validation Study

Category	n	Mean GDS	Min GDS	95% CI
Hull-interior	21	1.000	1.000	[1.000, 1.000]
Outside-hull	54	0.910	0.550	[0.885, 0.936]
Foreign-element	15	0.690	0.358	[0.594, 0.786]

The 5-tier RMSE classifier achieves 0% false-negative rate (0 of 21 hull-interior cases missed) and 3% false-positive rate (2 of 69 non-hull cases — both have RMSE < σ_instrument and are analytically indistinguishable from exact matches). The old GDS>0.97 threshold had a 43% false-positive rate; the new absolute RMSE classifier eliminates this problem.

12.3 Computational Self-Consistency Check

Two blends were constructed in silico from BOTTLE database CHNS measurements (Cuthbertson et al. 2024, CC BY-NC 3.0). Case 1 (30% PET / 50% PP / 20% Nylon 66): EBS recovered fractions within 1.2 pp of known values; RMSE = 0.16 pp (Noise-floor tier). Case 2 (40% HDPE / 35% PET / 25% PS, with PS absent from default source set): EBS returned a gracefully degraded result with RMSE = 1.40 pp (Acceptable tier) and $\chi^2 = 13.1$ ($p=0.003$), correctly flagging the poor fit.

12.4 Platform Testing

OS	Browser	Version	Result
Windows 11	Chrome	≥80	Full functionality
Windows 11	Edge	≥80	Full functionality
Windows 11	Firefox	≥78	Full functionality
macOS 13	Safari	≥14	Full functionality
Ubuntu 22.04	Chrome	≥80	Full functionality
iOS 16	Safari (mobile)	≥14	Solver functional; Word and SVG export unavailable (iOS download restriction)

12.5 Design Notes

- Population RMSE: $RMSE = \sqrt{(SSR/n)} \times 100$, dividing by n rather than n−1 (Bessel correction). Consistent with σ assumptions in χ^2. For n = 4–6 elements the difference is less than 15% of the RMSE value.
- Si atomic weight: EBS uses Si = 28.085 g/mol (IUPAC 2021). Earlier releases used 28.086; the 0.001 g/mol difference corresponds to a sub-0.001 pp elemental fraction error — negligible.
- NNLS exactness threshold: $SSR < 1\times10^{-8}$ used consistently throughout — status bar, Word export, Feasibility tool, Solution Space Explorer, and Leave-One-Out. Corresponds to per-element residual of $\sim1\times10^{-4}$, below elemental analysis instrument precision.

13. Troubleshooting

13.1 The file will not open when I double-click it

This is almost always a browser association issue, not a problem with the file. Right-click the file and choose Open with, then select Chrome, Edge, or Firefox. On macOS, right-click → Open With → Other... → navigate to Applications and choose your browser. Do not try to open the file from inside a zip archive — extract it first.

13.2 The page appears blank or shows only a white screen

Some browsers block local HTML files from running JavaScript. Try a different browser — Chrome and Edge have the fewest restrictions. As a last resort, serve the file locally: open a terminal in the folder containing the file and run:

```
python -m http.server 8080
```

Then open http://localhost:8080/molecular_solver_v9_0_2.html in your browser.

13.3 The formula I typed is not recognised

The formula parser reads standard Hill notation (C first, H second, then other elements alphabetically). Common mistakes:

- Case sensitivity: Write Cl not CL or cl; write Na not NA. The parser tries two-letter symbols first, so C3H6Si1 is read as C, H, Si — not C, H, S, I.
- No spaces or punctuation: Write C6H12O6, not C6 H12 O6.
- Only integer subscripts: Fractional subscripts (e.g. C4.78) are silently ignored. Convert tissue-reference weight fractions to nearest-integer pseudo-formulas before entry.
- Unsupported elements: Only the 20 tracked elements are recognised. Any other element symbol is silently ignored, giving a wrong molecular weight.

13.4 The solver shows Best-fit only — why no exact match?

An exact match requires the target to lie inside the convex hull of the source composition vectors. This fails when:

- A required element is missing from all sources (foreign element — shown in Composition Chart).
- The pinned fraction sum exceeds 100% (see §2.5 — proven infeasible regardless of other sources).
- The target composition is outside the range spanned by the sources.

Use the Substitution Recommender (§8.4) to find a better source polymer, or Exhaustive Search (§8.9) to find the globally optimal combination.

13.5 The Exhaustive Search is very slow

The search evaluates all C(pool, n) combinations. For n=5 with 55 eligible entries (Choline Chloride target), this is 3,478,761 combinations — ~58 minutes. Use n=3 (<1 s) as an effective first pass for most analyses. The search runs in a Web Worker so the UI remains responsive.

13.6 The exported Word report will not open

The report is standards-compliant OOXML tested with Microsoft Word 2016+, LibreOffice Writer 7+, and Google Docs. If charts appear as blank boxes, your word processor may not support inline SVG — use Microsoft Word or LibreOffice Writer. If you receive a "file format not supported" error, ensure the file was saved with the .docx extension and was not renamed by the browser.

13.7 Results changed unexpectedly after I changed a molecule

EBS re-runs the solver instantly whenever any molecule is changed — this is by design. Every result panel, chart, and advanced tool reflects the current state. Export a Word report before making changes to capture a snapshot.

14. Glossary

Terms are listed alphabetically.

Term	Definition
Active set (Z)	In the Lawson-Hanson NNLS algorithm, the set of source-polymer indices whose mixing fraction is currently held at zero. At convergence, all active-set variables satisfy KKT complementary slackness. Contrast with Passive set.
Atomic weight (AW)	Standard atomic weight of an element in g/mol, from IUPAC 2021 recommendations (Prohaska et al. 2022). Used to convert atom counts to mass contributions.
Best-fit blend	Solver output when no exact solution exists: the mixing-fraction vector that minimises $\|\|Af - b\|\|^2$ subject to $f \geq 0$ and $\Sigma f = 1$, even though the residual is non-zero. Indicated by non-Exact status bar.
Convex hull	The smallest convex set containing all source composition vectors. A target lies inside the hull if and only if an exact non-negative blend exists.
Diagnostic Advisor	Result-panel element integrating GDS tier and χ^2 outcome into a single actionable recommendation. Displayed after every solve. Resolves metric conflicts automatically.
Degrees of freedom (df)	In the χ^2 test, df = (number of active elements) − (number of sources with non-zero fraction), clamped to ≥1. For df ≤ 2, treat p-values as approximate indicators only.
Elemental weight fraction (w_x)	Fraction of a molecule's total mass contributed by element X: $w_x = AW(X) \times count(X) / M$. All elemental weight fractions for a molecule sum to 1.0.
Exact solution	A mixing-fraction vector f satisfying Af = b exactly (SSR < 1×10^{-8}), with all fractions non-negative and summing to 1. Exists if and only if the target lies inside the convex hull.
Foreign element	An element present in the target molecule but absent from all selected source materials. Identified in the Composition Chart (View 6) with annotation "Not in sources".
GDS (Global Decomposability Score)	GDS = $\max(0, 1 - RMSE/\|\|b\|\|_2)$. Dimensionless scalar [0,1]; higher is better. Use for trend monitoring; use the 5-tier RMSE classifier for pass/fail decisions.
Hill notation	Convention for writing molecular formulae: carbon first, hydrogen second, then all other elements alphabetically. EBS accepts formulae in Hill notation.
KKT conditions	Karush-Kuhn-Tucker optimality conditions for the NNLS problem. The Lawson-Hanson algorithm terminates when these are satisfied.
Lawson-Hanson algorithm	Active-set method for solving Non-Negative Least Squares problems. Ref: Lawson, C.L. & Hanson, R.J. (1995). Solving Least Squares Problems. SIAM.

Term	Definition
Mass fraction (f_i)	Fraction of the total blend mass contributed by source polymer i. All mass fractions satisfy $f_i \geq 0$ and $\Sigma f_i = 1$.
NNLS	Non-Negative Least Squares: minimise $\|\|Af - b\|\|^2$ subject to $f \geq 0$.
Passive set (P)	In the Lawson-Hanson algorithm, the set of source indices currently allowed to be positive. The unconstrained LS sub-problem is solved over P at each inner iteration.
Penalty row	Extra row appended to A and b with value $\lambda = 10$. Converts the equality constraint $\Sigma f_i = 1$ into an unconstrained NNLS problem.
Pinned fraction	Mass fraction of a source polymer uniquely determined because it is the only supplier of a particular element. If pinned fractions sum to >1, the problem is proven infeasible.
RMSE	Root Mean Square Error: $RMSE = \sqrt{(SSR/m)} \times 100$ pp, where m is the number of active elements. Primary basis for the 5-tier classifier.
Solution Space Explorer	Diagnostic tool characterising the algebraic solution space. For exact solutions, reports per-source fraction ranges across all equivalent blends (polytope vertices). For unique solutions, identifies equivalent source substitutions.
SSR	Sum of Squared Residuals: $SSR = \|\|A\hat{f} - b\|\|^2 = \Sigma_e(achieved_e - target_e)^2$. $SSR < 1\times10^{-8}$ is the threshold for declaring an exact solution.
χ^2 (Chi-squared)	$\chi^2 = \Sigma_e(achieved_e - target_e)^2/\sigma_e^2$. Element-specific σ: C/H/N=0.003, O/S/P/Si/Na/K/Mg/Ca/Fe/Al/Zn/B=0.004, Cl/F/Br/I/Se=0.005 (weight fractions). df = m − nactive, clamped to ≥1.

15. Repository and Citation

File / Resource	Description
molecular_solver_v9_0_2.html	Complete self-contained solver application. Open in any modern browser — no installation. Contains full NNLS solver, 73-polymer database, 16 analysis tools, and Word report export engine.
Repository URL	https://github.com/pemason437-svg/Elemental-Blend-Solver
Zenodo DOI	10.5281/zenodo.19263118
Bug reports / Features	https://github.com/pemason437-svg/Elemental-Blend-Solver/issues
README.md	Quick-start guide, feature summary, mathematical background overview, validation summary, and contact details.
LICENSE.txt	MIT License. Copyright © 2026 Paul Mason. Permits free use, modification, and distribution with attribution.
CITATION.cff	Machine-readable citation metadata in Citation File Format. GitHub uses this file to populate the "Cite this repository" button.
paper/ folder	Journal paper source files (LaTeX and bibliography) for manuscripts under review.

To clone the repository: git clone https://github.com/pemason437-svg/Elemental-Blend-Solver.git

To cite EBS in academic work, use the Zenodo DOI above or click the "Cite this repository" button on the GitHub page. A companion paper describing the mathematical framework, software architecture, and validation results is currently under review.